

ECE148: Signal Processing Applications

Assignment 2: Signal Aliasing (Part B)

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1 Fourier Spectrum

A simple coherent signal $f(t)$ is in the form $f(t) = e^{j\omega_x t}$. The Fourier spectrum of this signal consists of one single peak at $\omega = \omega_x$. This coherent waveform is first sampled, with sample spacing Δt . Subsequently, it is reconstructed through a low-pass filter with cutoff frequencies at $\pm \frac{\omega_0}{2}$ where $\omega_0 = \frac{2\pi}{\Delta t}$. Then, the reconstructed signal is in the form $g(t) = e^{j\omega_y t}$, and its Fourier spectrum shows a single peak at $\omega = \omega_y$. If the sampling and reconstruction are conducted adequately, the frequency remains unchanged, $\omega_y = \omega_x$. When aliasing occurs, the resulting frequency will be different. The conversion of frequency from ω_x to ω_y is a mapping process.

(a) Mapping correspondence between ω_x and ω_y

When the signal is sampled with sampling interval Δt , the angular sampling frequency is:

$$\omega_0 = \frac{2\pi}{\Delta t}$$

Sampling causes the spectrum to repeat every ω_0 , so any original frequency ω_x is indistinguishable from:

$$\omega_x + m\omega_0, \quad k \in \mathbb{Z}$$

The low-pass reconstruction filter keeps only frequencies in $[-\frac{\omega_0}{2}, \frac{\omega_0}{2}]$, so the reconstructed frequency ω_y is the aliased version of ω_x folded into this interval:

$$\omega_y = \omega_x - m\omega_0$$

where m is chosen such that $\omega_y \in [-\frac{\omega_0}{2}, \frac{\omega_0}{2}]$.

(b) The (ω_x, ω_y) Correspondence Curve

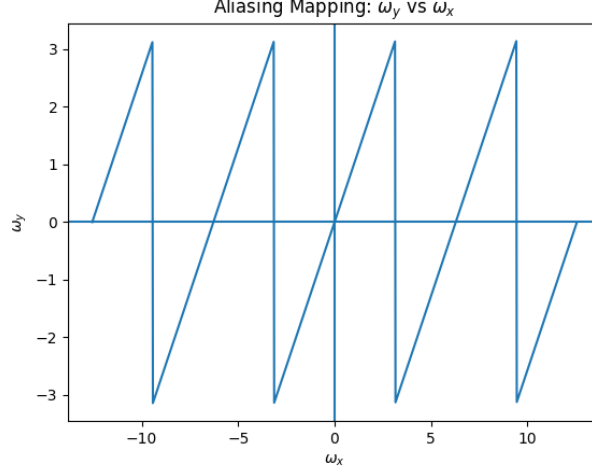


Figure 1: Mapping correspondence between ω_x and ω_y (Continuous Mapping)

2 Discrete-Time Fourier (DTFT) Spectrum

Using the same coherent waveform $f(t) = e^{j\omega_x t}$, we sample the waveform with sample spacing Δt to form a discrete sequence $f(n\Delta t) = e^{j\omega_x n\Delta t}$ to compute the DTFT spectrum of the sequence:

$$F(e^{j\theta}) = DTFT\{f(n\Delta t)\} = \sum_{n=-\infty}^{\infty} f(n\Delta t)e^{-jn\theta}$$

Corresponding to the peak at $\omega = \omega_x$, the DTFT spectrum also has a peak at $\theta = \theta_y$. Defining $\theta_x = \omega_x \Delta t$, the sequence becomes $f[n] = e^{jn\theta_x}$. The DTFT, therefore, is:

$$F(e^{j\theta}) = \sum_{n=-\infty}^{\infty} e^{jn\theta_x} e^{-jn\theta} = \sum_{n=-\infty}^{\infty} e^{jn(\theta_x - \theta)}$$

This produces impulses at $\theta = \theta_x + 2\pi m$, $m \in \mathbb{Z}$. The DTFT frequency is always restricted to $[-\pi, \pi]$, so the observed frequency θ_y is the wrapped version of θ_x :

$$\theta_y = \theta_x - 2\pi m$$

where $\theta_y \in [-\pi, \pi]$. Substituting $\theta_x = \omega_x \Delta t$:

$$\theta_y = \omega_x \Delta t - 2\pi m$$

with m such that $\theta_y \in [-\pi, \pi]$.

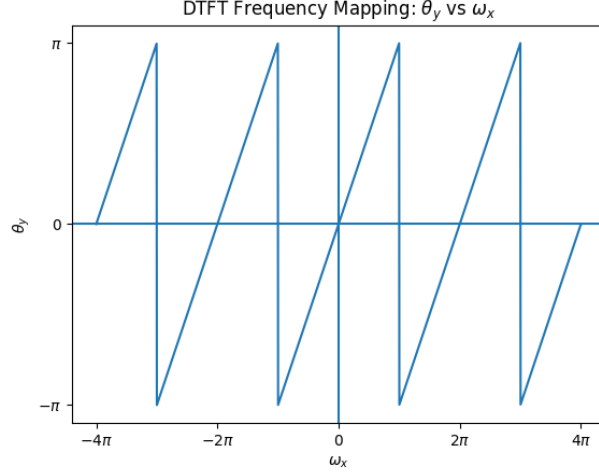


Figure 2: Mapping correspondence between ω_x and θ_y (Digital Mapping)

3 DFT Spectrum

As before, we define:

$$f[n] = e^{jn\theta_x}, \quad \theta_x = \omega_x \Delta t$$

The N -point DFT is:

$$F[k] = \sum_{n=0}^{N-1} f[n] e^{-jn \frac{2\pi}{N} k}$$

Substituting:

$$F[k] = \sum_{n=0}^{N-1} e^{jn\theta_x} e^{-jn \frac{2\pi}{N} k} = \sum_{n=0}^{N-1} e^{jn(\theta_x - \frac{2\pi}{N} k)}$$

The DFT has a peak when $\theta_x = \frac{2\pi}{N} k$, where $\theta_x \in [0, 2\pi]$ and $k \in [0, 1, \dots, N-1]$. Mapping continuous frequency ω_x to discrete index k_y :

$$\omega_x \Delta t = \frac{2\pi}{N} k_y$$

$$k_y = \frac{\Delta t}{2\pi} N \omega_x$$

$$k_y = N \frac{\omega_x}{\omega_0} \pmod{N}$$

for $k_y \in [0, N-1]$.

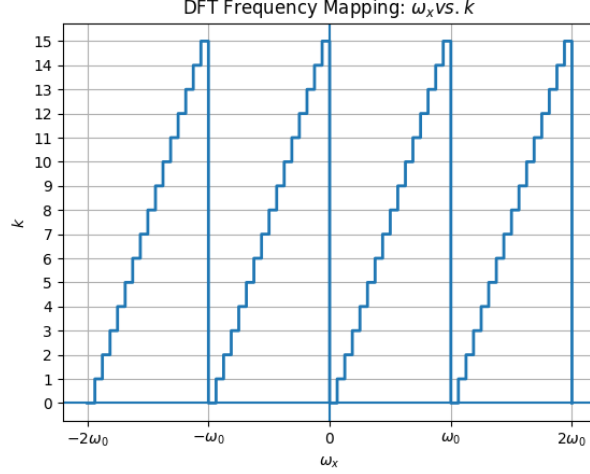


Figure 3: Mapping correspondence between ω_x and k (Bin Mapping)

4 Summary

Fourier Spectrum (Continuous Mapping)

Given an input $f(t) = e^{j\omega_x t}$, the spectrum will have a single impulse at $\omega = \omega_x$. After sampling with interval Δt , the spectrum becomes periodic with period $\omega_0 = \frac{2\pi}{\Delta t}$. Following reconstruction, the observed frequency is:

$$\omega_y = \omega_x - m\omega_0$$

Thus, frequencies are folded into the interval $[-\frac{\omega_0}{2}, \frac{\omega_0}{2}]$

DTFT Spectrum (Digital Mapping)

Sampling produces the discrete-time signal $f[n] = e^{jn\omega_x t}$, where the discrete-time frequency is defined as $\theta = \omega_x \Delta t$. The DTFT spectrum is periodic with period 2π , and the frequency mapping is:

$$\theta_y = \omega_x \Delta t - 2\pi m$$

Here, the frequency is wrapped into $\theta \in [-\pi, \pi]$.

DFT Spectrum (Bin Mapping)

The N -point DFT samples the DTFT at $\theta_x = \frac{2\pi}{N}k_y$. The corresponding k_y for a given frequency is $k_y = \frac{N}{2\pi}\theta_x$. Substituting $\theta_x = \omega_x\Delta t$, we obtain:

$$k_y = N \frac{\omega_x}{\omega_0} \pmod{N}$$

for $k_y \in [0, N - 1]$